

GLYCOGEN METABOLISM

Dr Balakrishnan. S, PhD.,
Senior Lecturer of Biochemistry

Objectives:

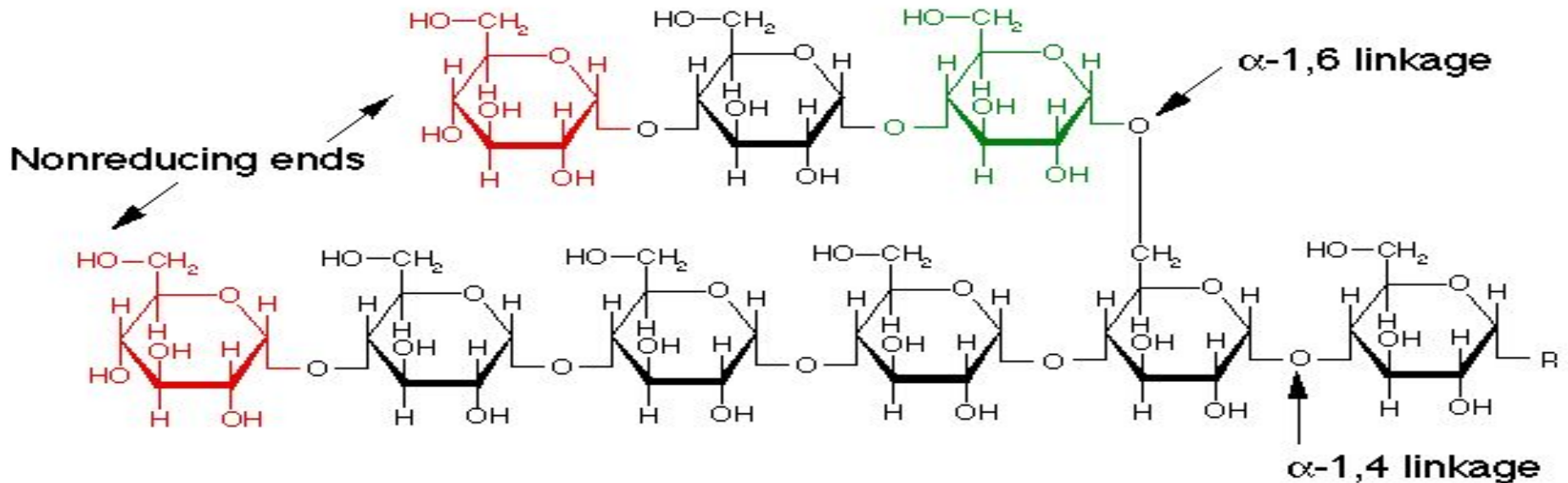
- At the end of this lecture, it is expected to;
 - Describe the Importance of Glycogen and its metabolism
 - Describe the pathways of Glycogenolysis and Glycogenesis
 - Differentiate Glycogenesis from Gluconeogenesis
 - Understand the regulations of Glycogen metabolism
 - Explain the hormonal controls of Glycogen metabolism
 - Describe the conditions of glycogen storage diseases

Biomedical Importance

- The reserve carbohydrate of the animal
- In a healthy adult in the fed state:
 - Liver contains about 400 g of glycogen, whereas muscle contains approximately 1200 g of glycogen.
- Maintenance of blood glucose:
 - Liver glycogen is largely concerned with storage and supply of glucose-1-P, which is converted to glucose particularly in between meals.
- Provision of energy:
 - Muscle glycogen is to act as readily available source of intermediates of glycolysis for provision of energy within the muscle itself.
 - Muscle glycogen cannot directly contribute to blood glucose level.
- Glycogen storage diseases (GSDs):
 - Inherited deficiency of enzymes in the pathway of glycogen metabolism produces certain inherited disorders called as GSDs.

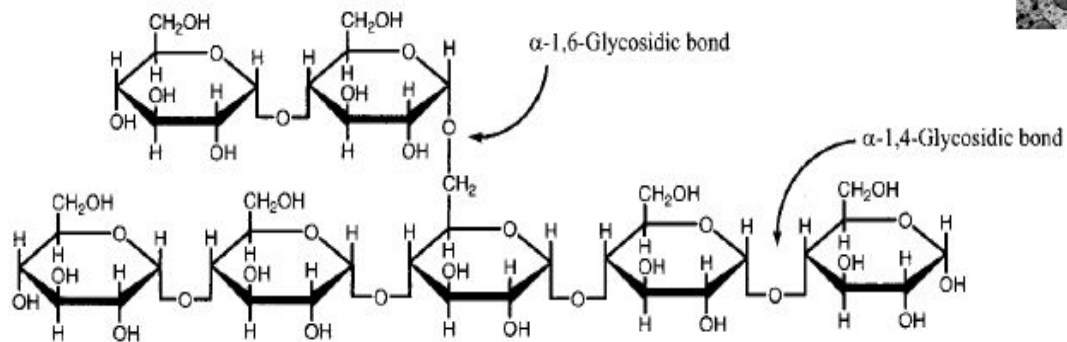
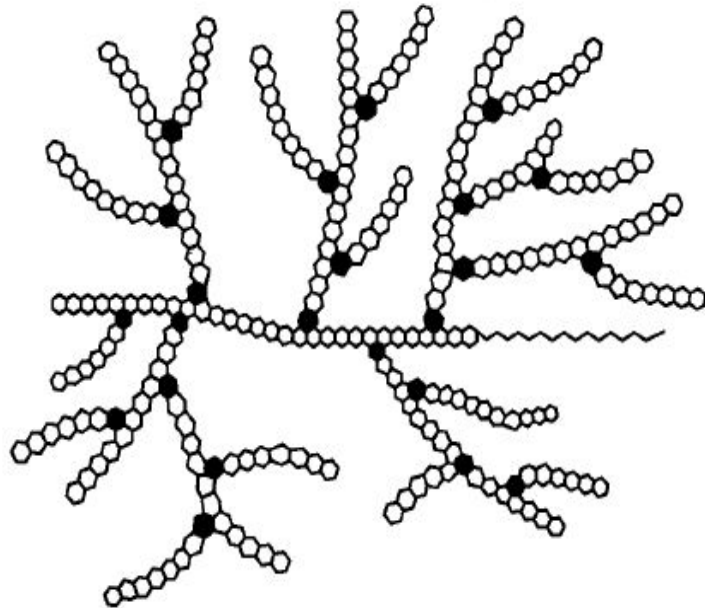
Glycogen:

- A highly branched polymer of glucose.
- Chains have glycosidic links α 1 $\rightarrow$ 4.
- Branches are linked α 1 $\rightarrow$ 6.

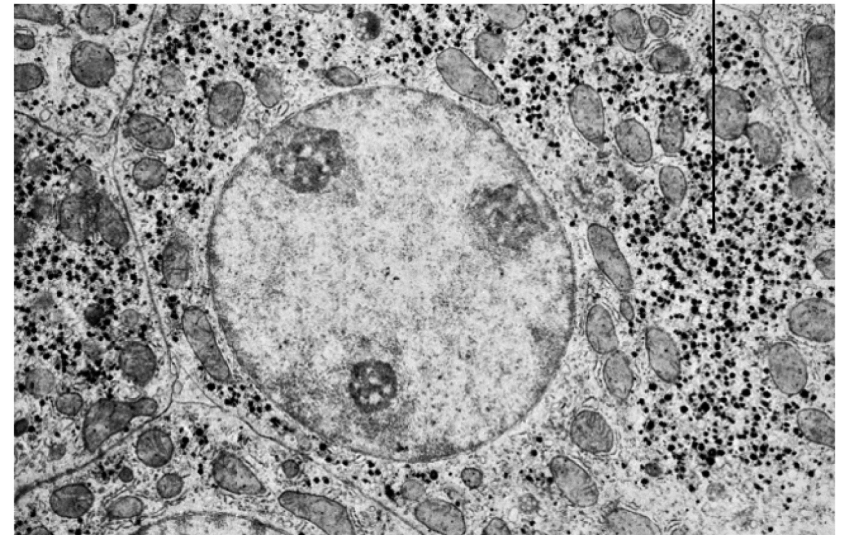


Structural features of glycogen

Structure of glycogen



Glycogen granules

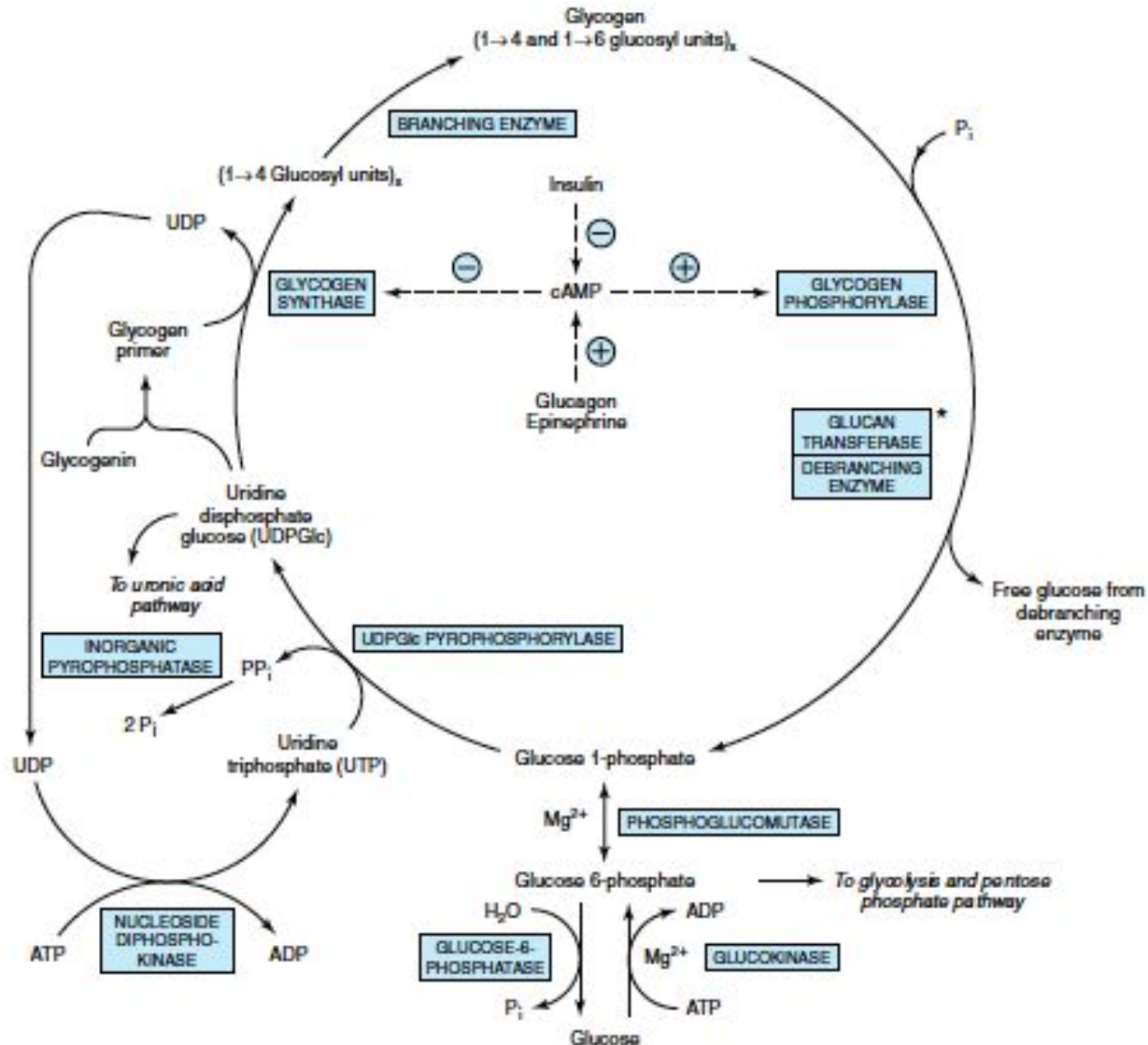


Liver Cell

GLYCOGEN METABOLISM

- ◆ Enough glucose and energy triggers synthesis of glycogen = **glycogenesis**
 - Synthesis of glycogen occurs during the fed state and is stimulated by **insulin**
- ◆ Glucose can be rapidly delivered to blood stream when needed upon degradation of glycogen = **glycogenolysis**
 - Glycogenolysis in liver occurs in the fasting state and is stimulated primarily by the hormone **glucagon**

Glycogen metabolism: overview



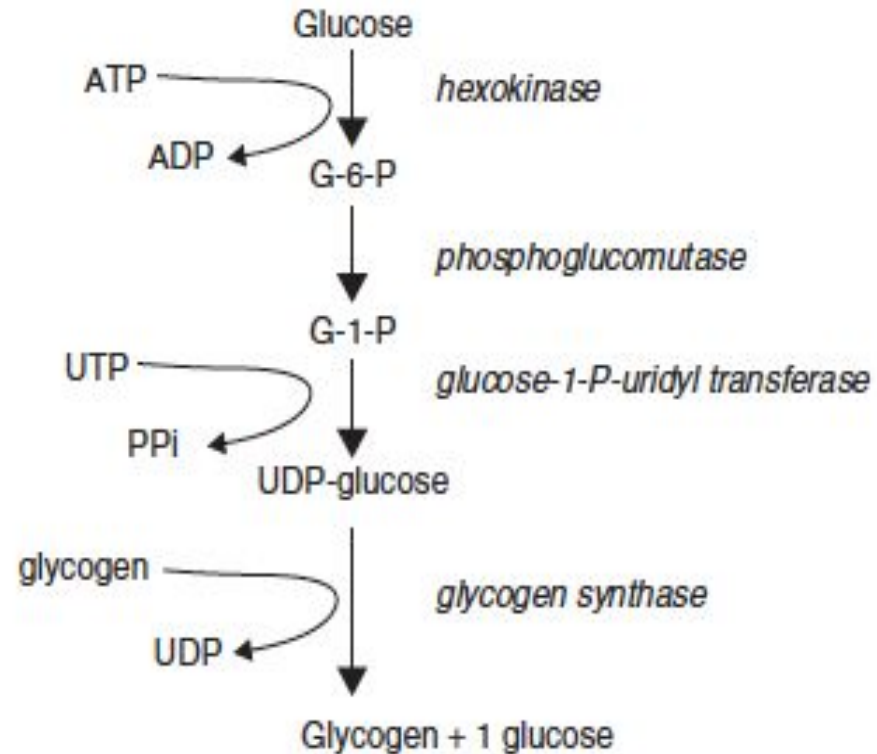
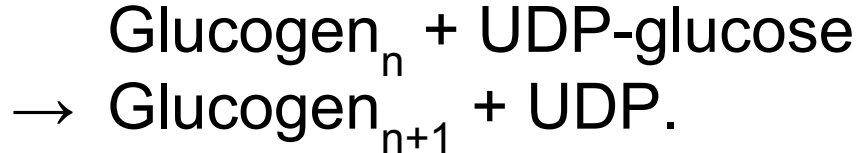
Glycogenesis:

The Synthesis of Glycogen
An Energy Consuming Pathway

Glycogenesis: overview

- Glycogen is synthesized via uridine diphosphate glucose (UDP – glucose).

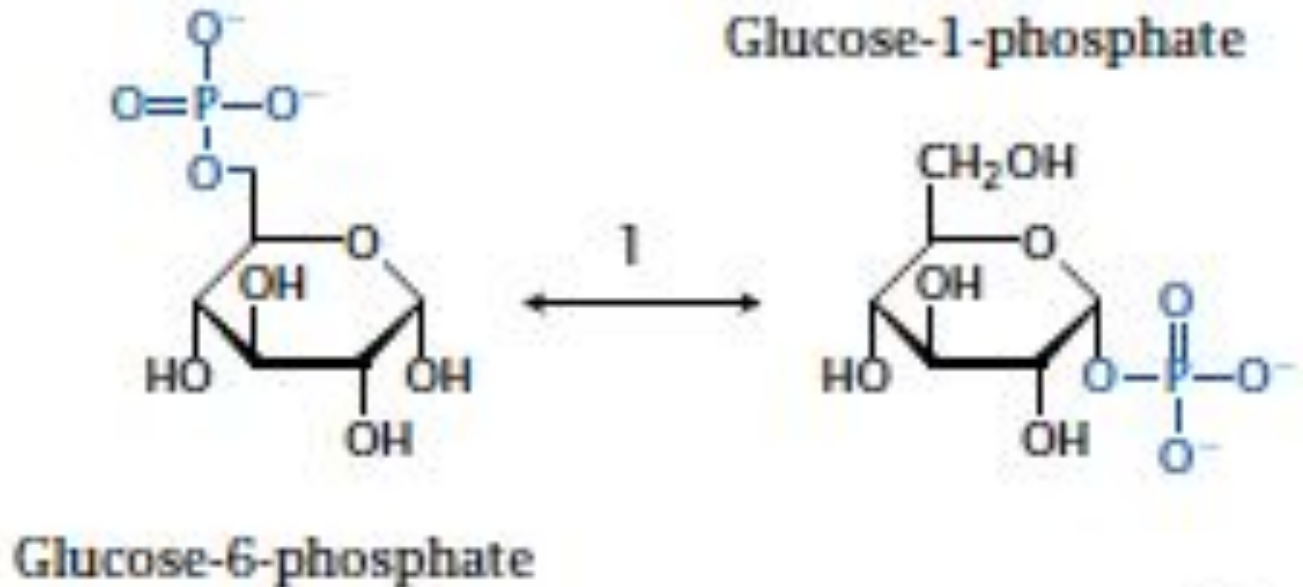
- Synthesis:



Glycogenesis steps:

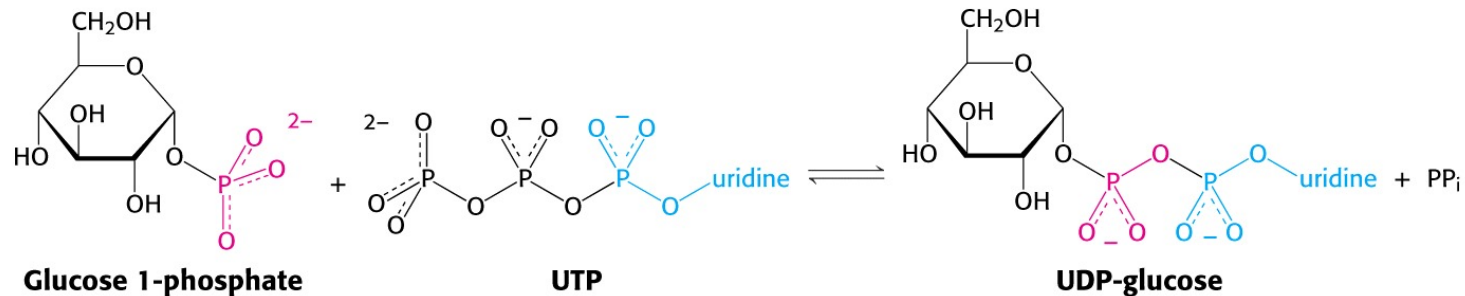
1. Synthesis of an activated precursor, UDP-glucose, by UTP:glucose-1-phosphate uridylyltransferase
2. Initiation of glycogen synthesis by glycogenin
3. Introduction of branches by branching enzyme
4. Chain elongation by glycogen synthase
5. Repeat steps 3 and 4

1. Conversion of glucose-6-phosphate to glucose-1-phosphate



- Enzyme = **phosphoglucomutase**
- Reversible reaction allows G1P conversion to G6P in glycogenolysis

2. Synthesis of Uridine Diphosphoglucose



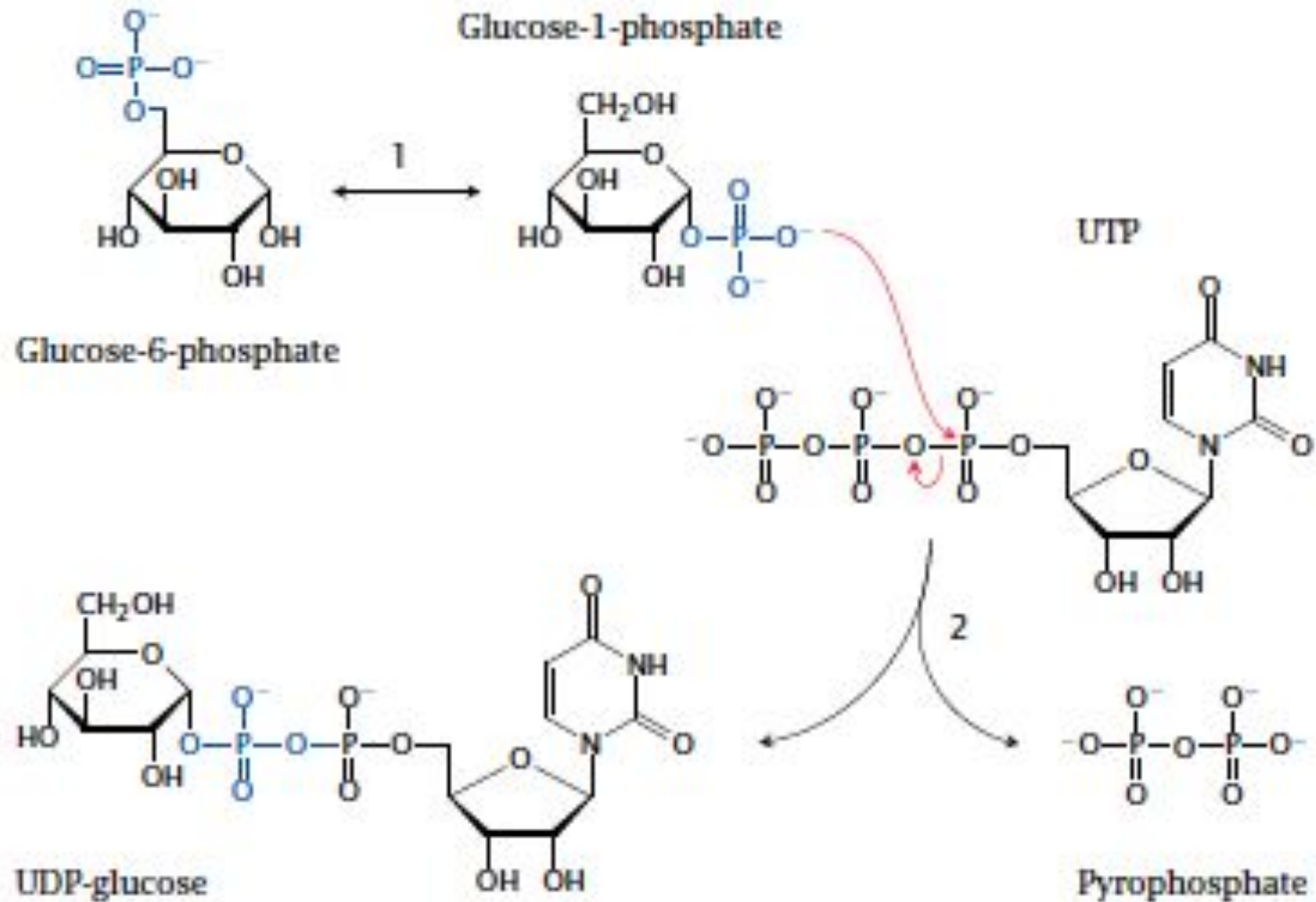
Then $PP_i \rightarrow 2 P_i$

By

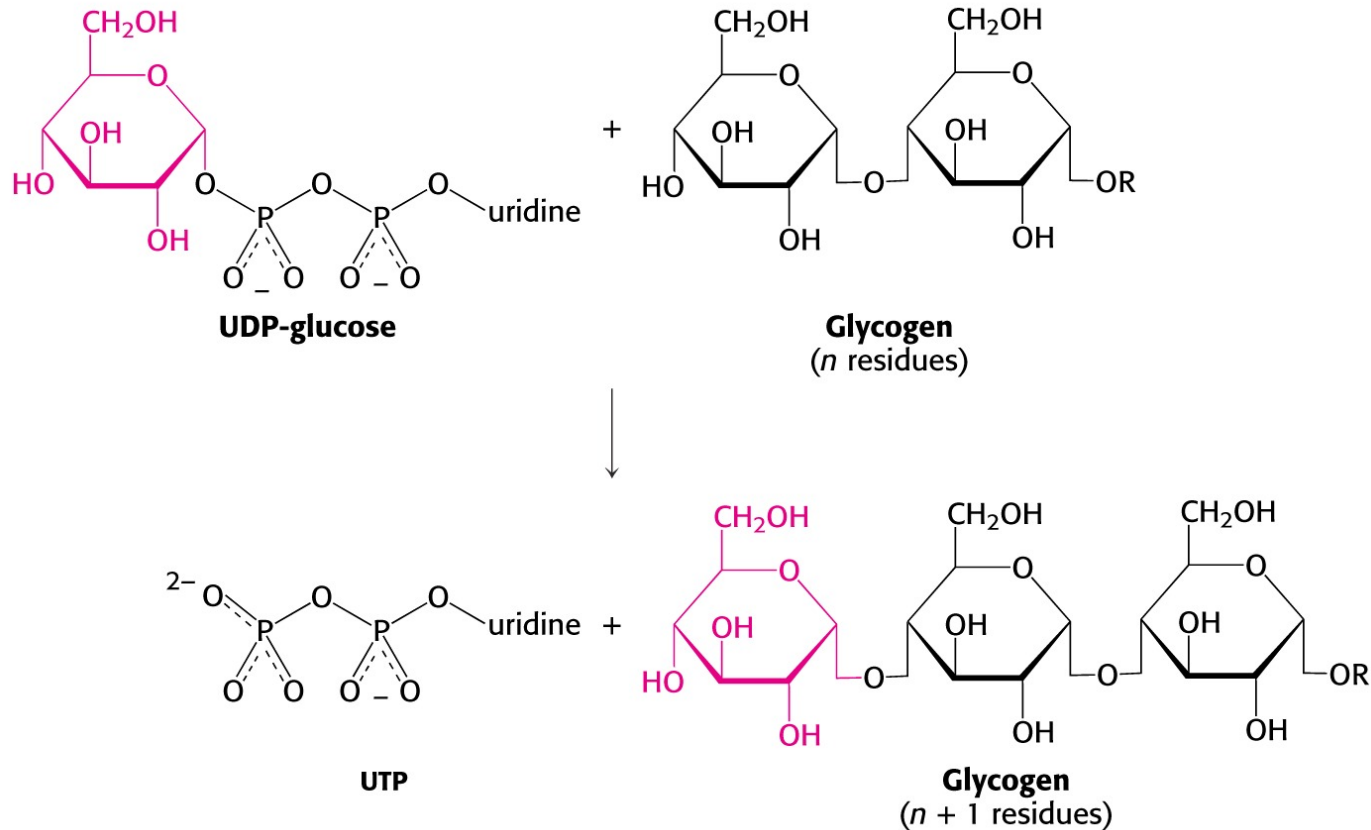
Pyrophosphatase

- Enzyme = **UDP-glucose pyrophosphorylase**
- UDP glucose is the activated form of glucose.
- Although the reaction is reversible the hydrolysis of the pyrophosphate pushes it to the right.

Synthesis of an activated precursor, UDP-glucose



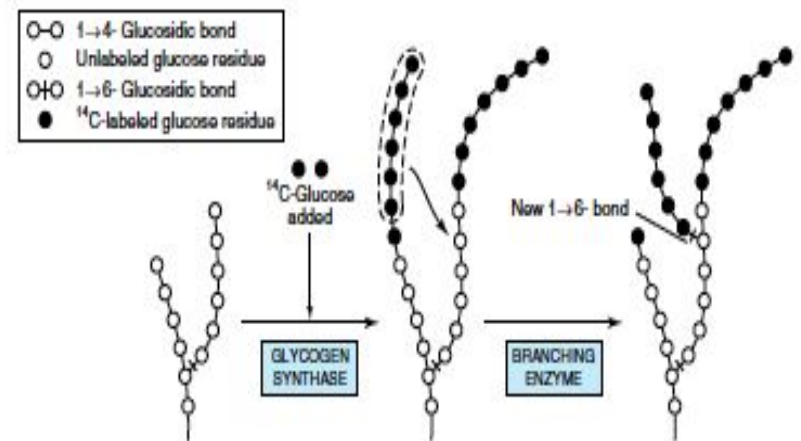
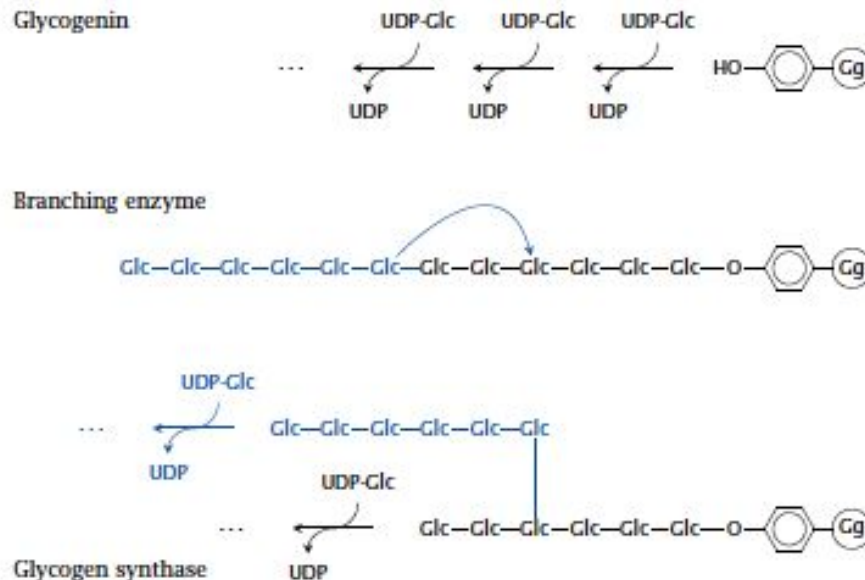
3. Glycogen synthesis



- Enzyme = **glycogen synthase**
- A primer of at least 4 units are required via glycogenin.
- Glucose is added to the non-reducing end.

4. Branching of glycogen

- Enzyme = **branching enzyme**
- Branching enzyme establishing a branch point:
 - When the chain has been lengthened to at least 11 glucose residues, transfers a part of the 1→4 chain (at least six glucose residues) to a neighboring chain to form a 1→6 linkage.
 - Makes a branch with an α (1→6) link and two ends to add glucose.

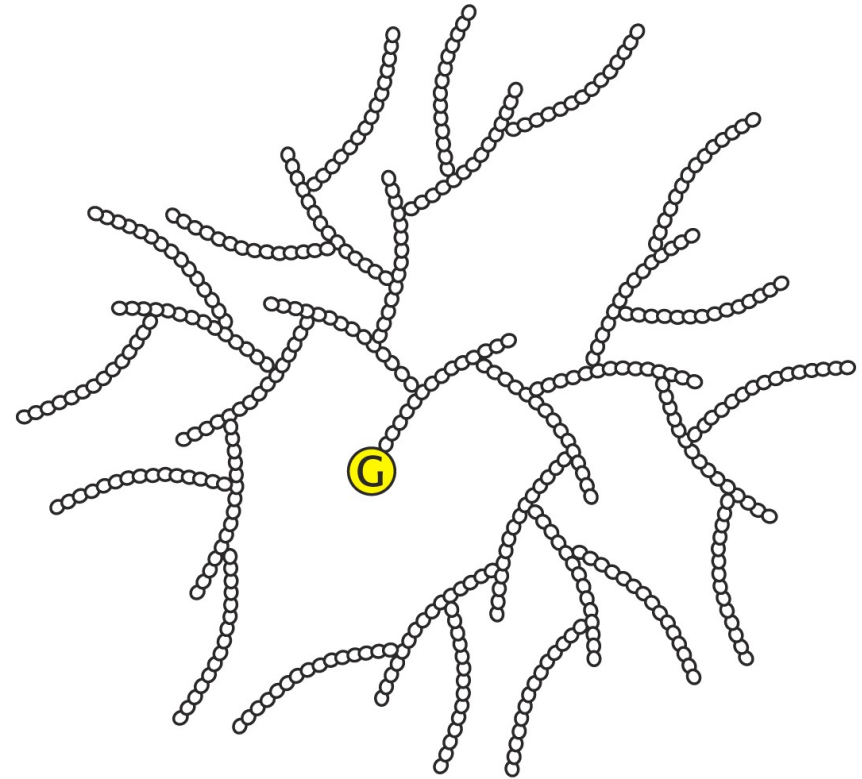


Branching enzyme forms α -1,6 linkages:

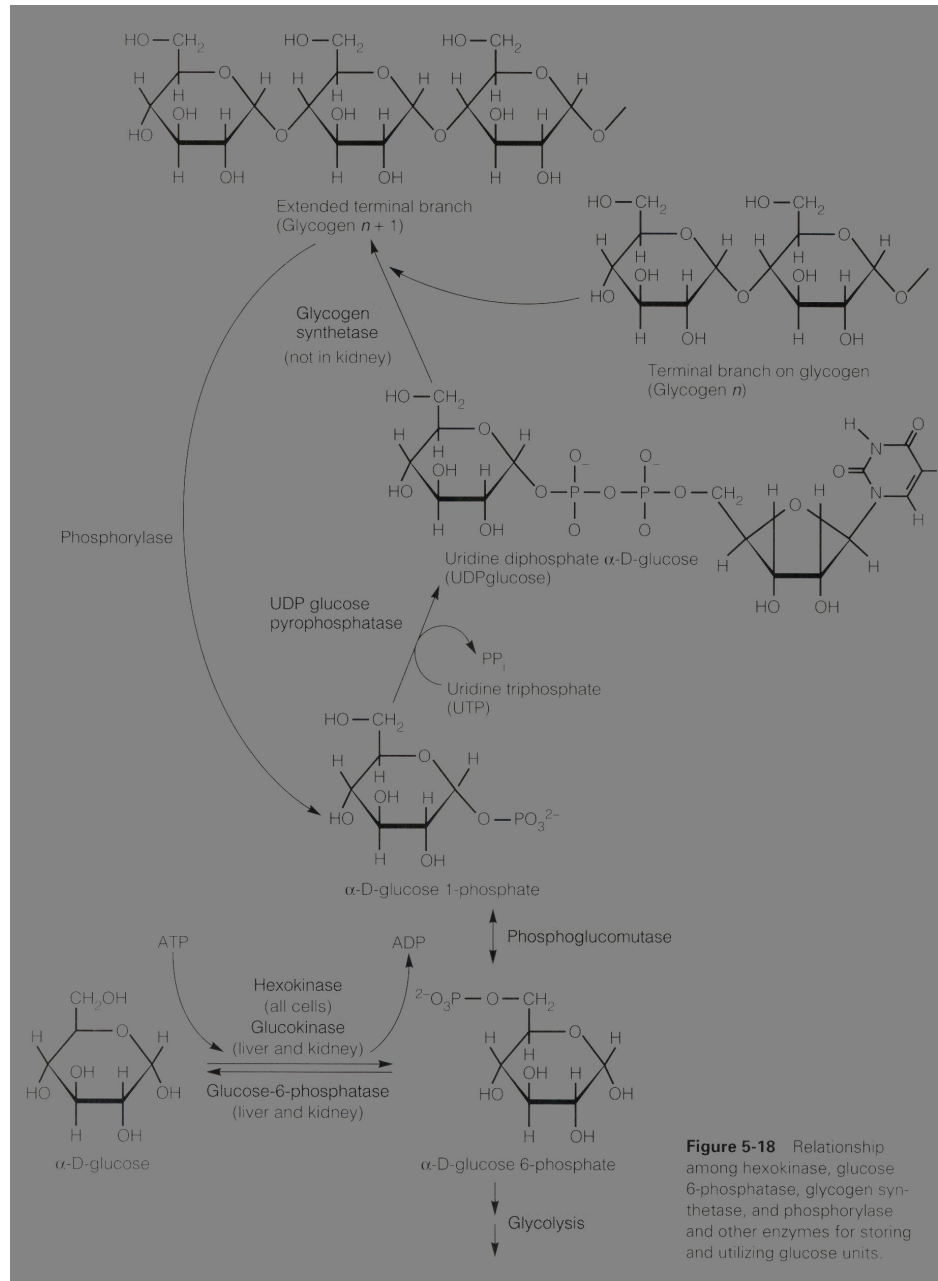
- The enzyme breaks the α -1,4 link and forms a α -1,6 link.

Advantages:

- Accelerates the rate of glucose release during degradation.
 - A large number of terminal residues are now available for glycogen phosphorylase
- Increases the solubility of glycogen.



Glycogenesis:



Brooks et al.

Summary of Glycogen synthesis

Glucose 6-P $\rightarrow$ glucose 1-P.

glucose 1-P + UTP $\rightarrow$ UDP-glucose + PP_i.

PP_i + H₂O $\rightarrow$ 2 P_i.

UDP-glucose + glycogen_n $\rightarrow$ glycogen_{n+1}.

UDP + ATP $\rightarrow$ UTP + ADP.

Net Reaction:

Glucose 6-P + ATP + glycogen_n + H₂O $\rightarrow$ glycogen_{n+1} + ADP + 2P_i

- Only one ATP is used to store one glucose residue in glycogen.

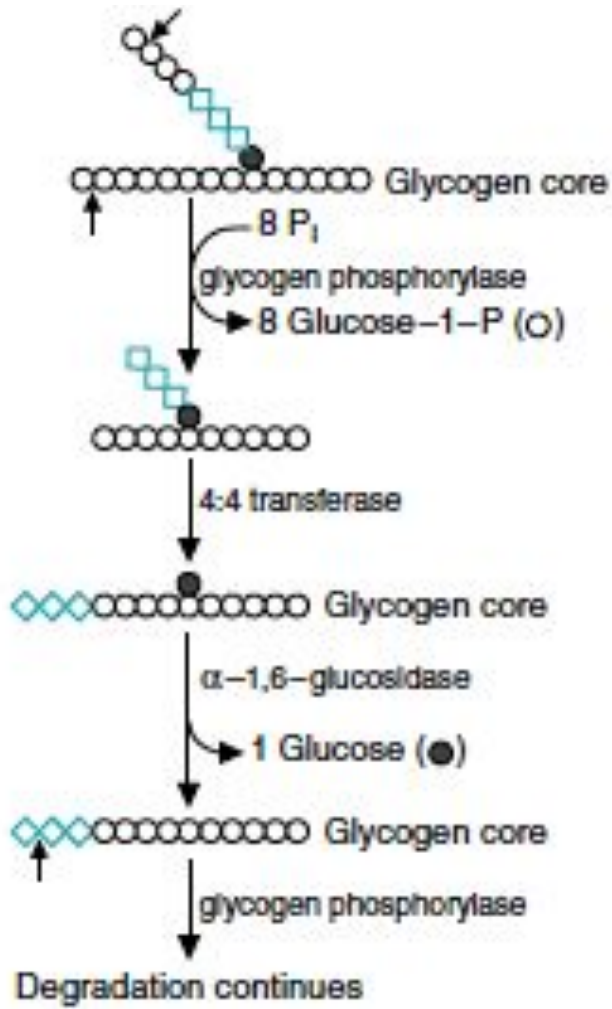
Glycogenesis Vs Gluconeogenesis

GLYCOGENESIS	GLUCONEOGENESIS
Synthesis of glycogen from glucose	Synthesis of glucose from non carbohydrate substrates
Stimulated by increased blood glucose level as in well fed state, insulin	Stimulated by low glucose levels like starvation, acidosis, glucagon and glucocorticoids
Inhibited by glucagon, epinephrine	Inhibited by insulin
First step is synthesis of UDPGlc	First step is Pyruvate
Key enzyme is Glycogen synthase	There are 4 key enzymes – Pyruvate carboxylase, Phosphoenol pyruvate carboxykinase, Fructose 1, 6 biphosphatase and Glucose 6 phosphatase
Occur in liver and muscle	Occur in liver and kidney

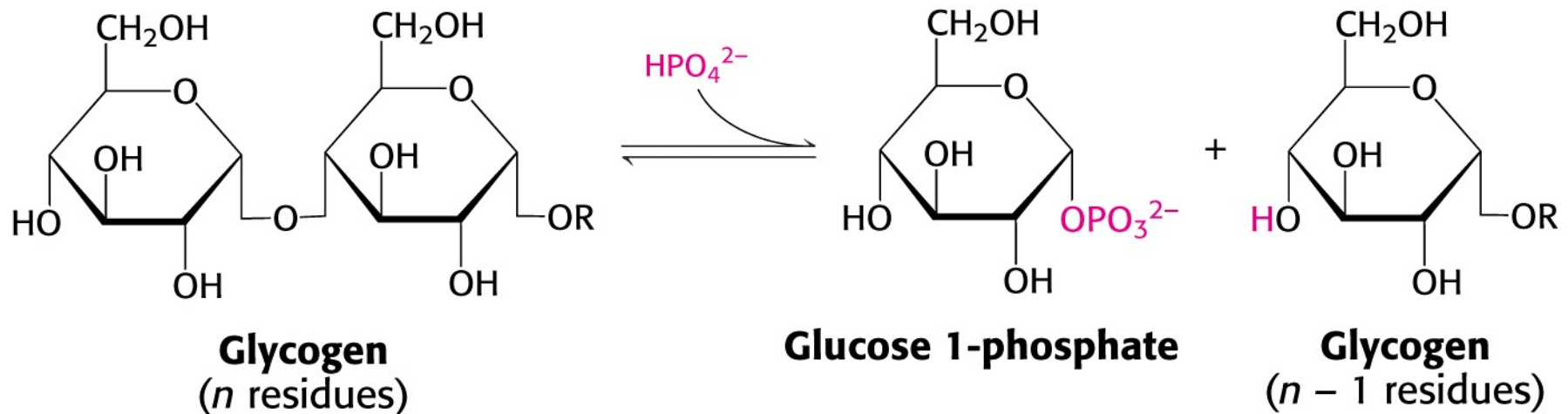
Glycogen Mobilization: **Glycogenolysis**

- Glycogenolysis is a catabolic process; the breakdown of glycogen to glucose units.
- The degradation of stored glycogen in liver and muscle constitutes glycogenolysis.
- It is not the reverse of glycogenesis but is a separate pathway.

Glycogenolysis: Overview

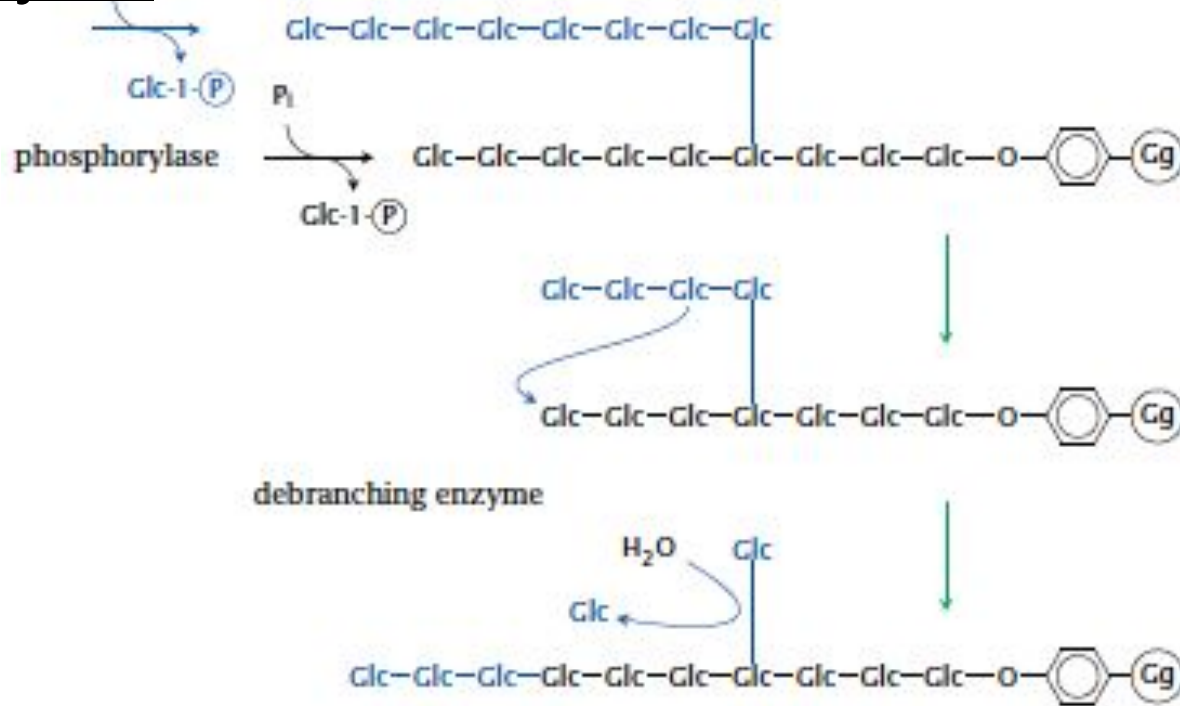


1. First step catalyzed by Glycogen phosphorylase



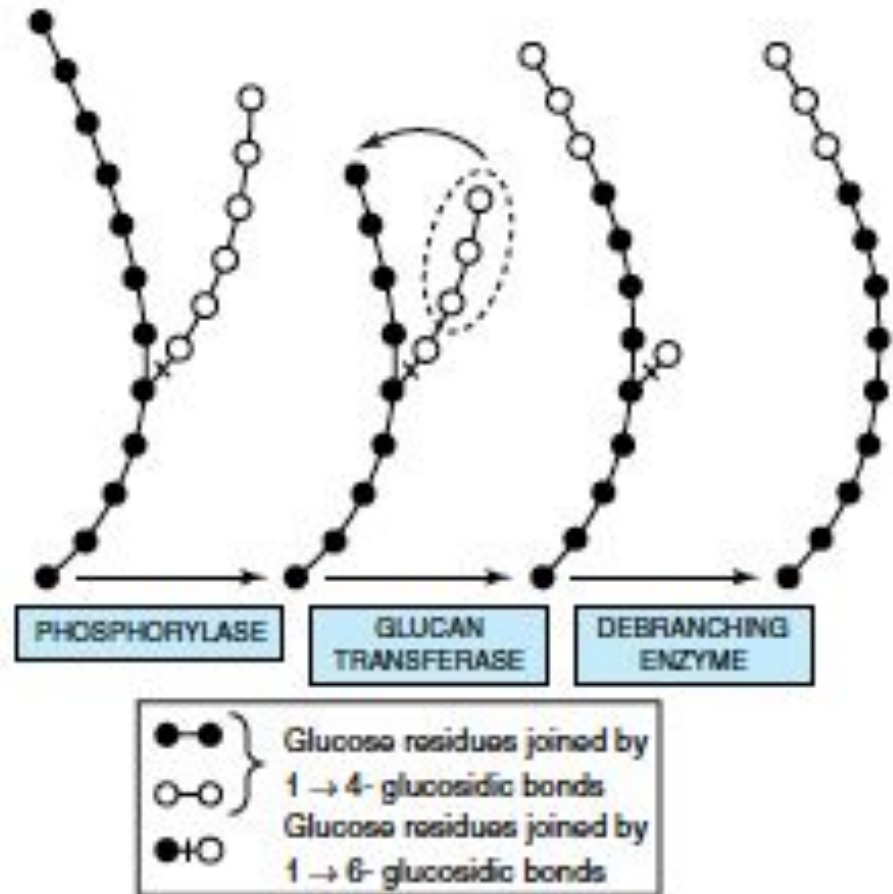
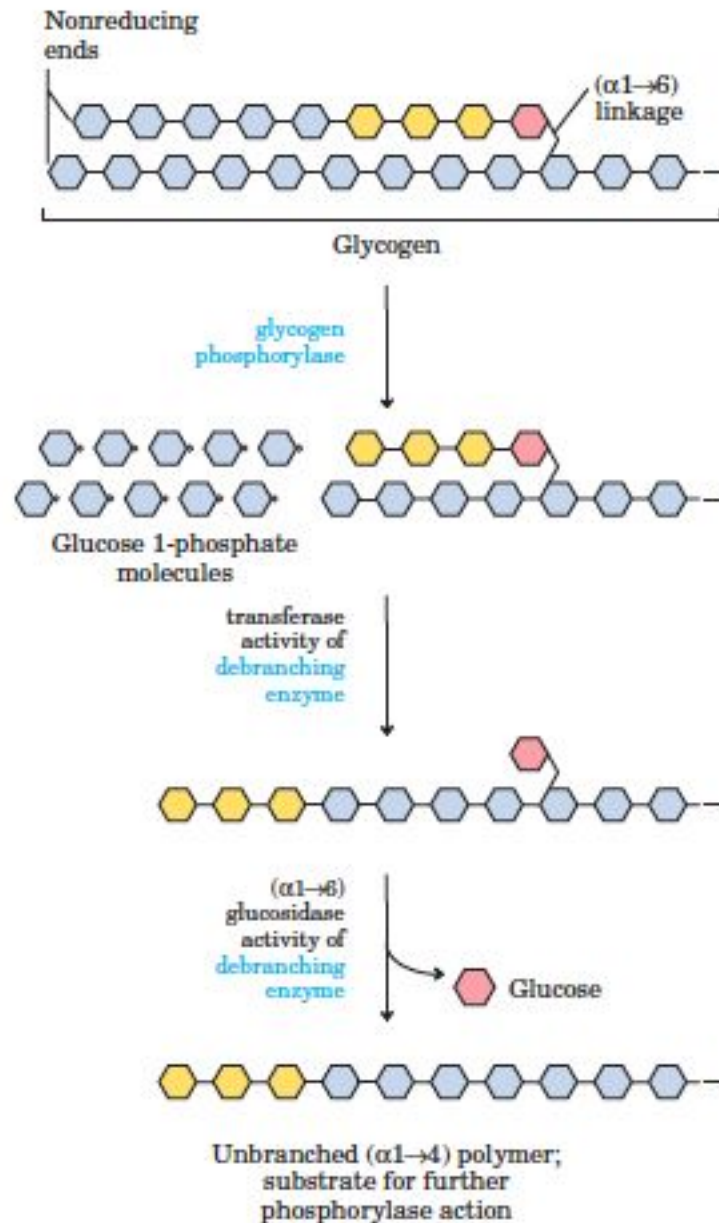
- Here, glycogen phosphorylase cleaves the bond at the 1 position by substitution of a phosphoryl group.
- It breaks down glucose polymer at α -1-4 linkages until 4 linked glucoses are left on the branch.
- Glycogen phosphorylase can be used as a marker enzyme to determine glycogen breakdown.

2. Second step catalyzed by debranching enzyme

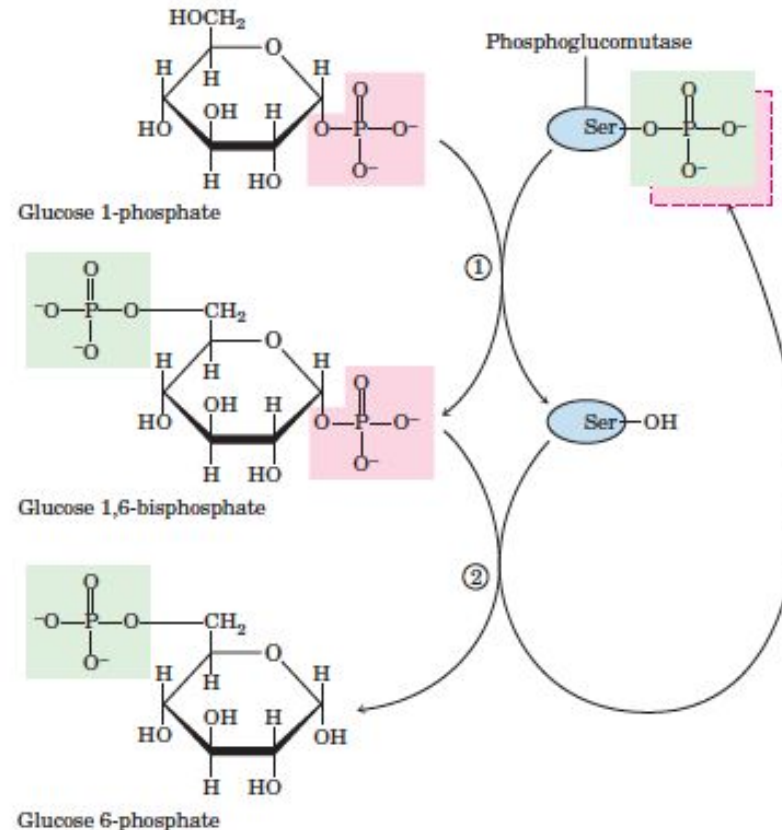


- Involves the debranching enzyme (4:4 transferase) that moves three remaining glucose units to another 1,4 end of glycogen.
- The debranching enzyme (α -1,6-glucosidase) hydrolysis the glucose attached as a 1,6-branched mono residue $\square$ giving one free glucose molecule.
- $\square$ This is the only case in which a glycogen metabolite is not glucose-1-phosphate.

Mechanism of debranching enzyme

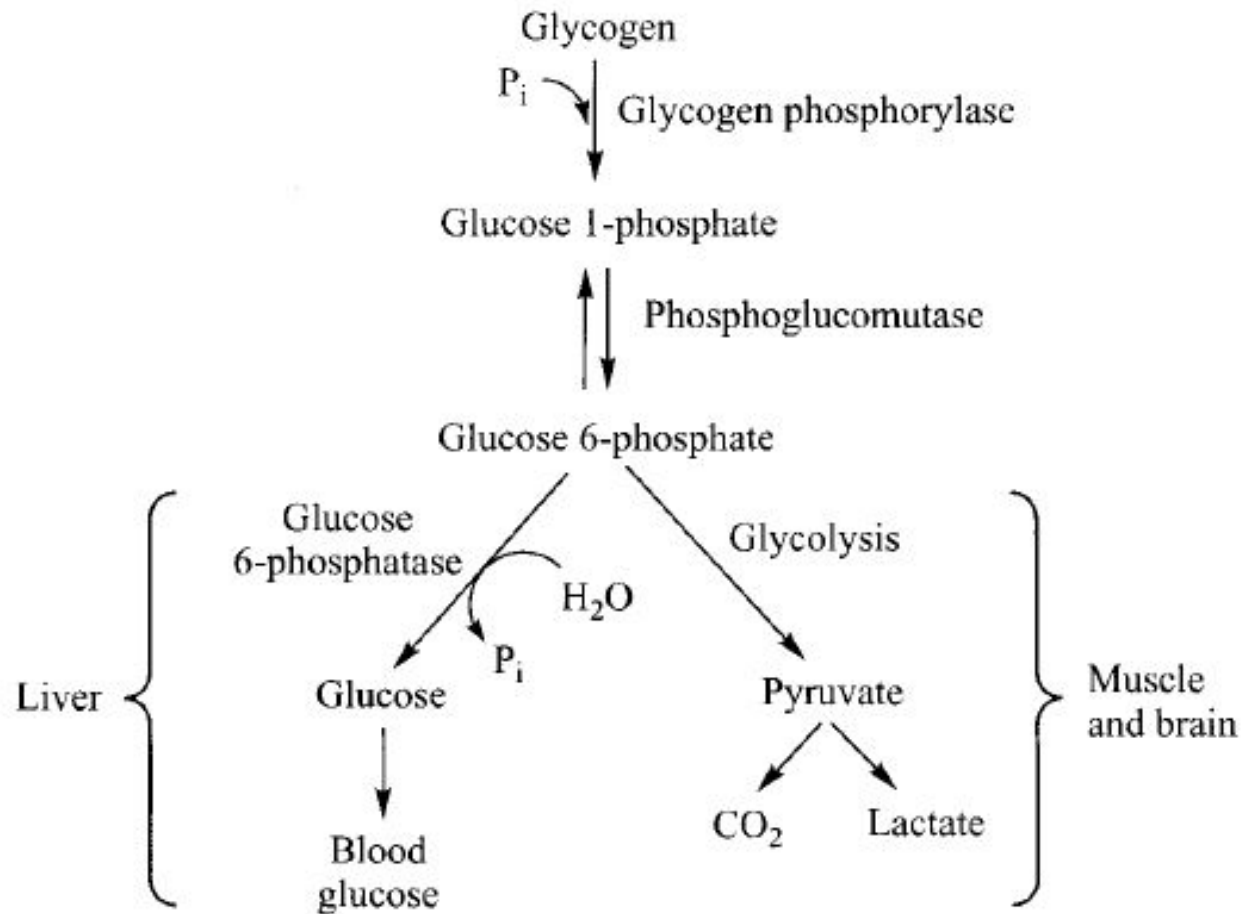


3. Third step catalyzed by phosphoglucomutase

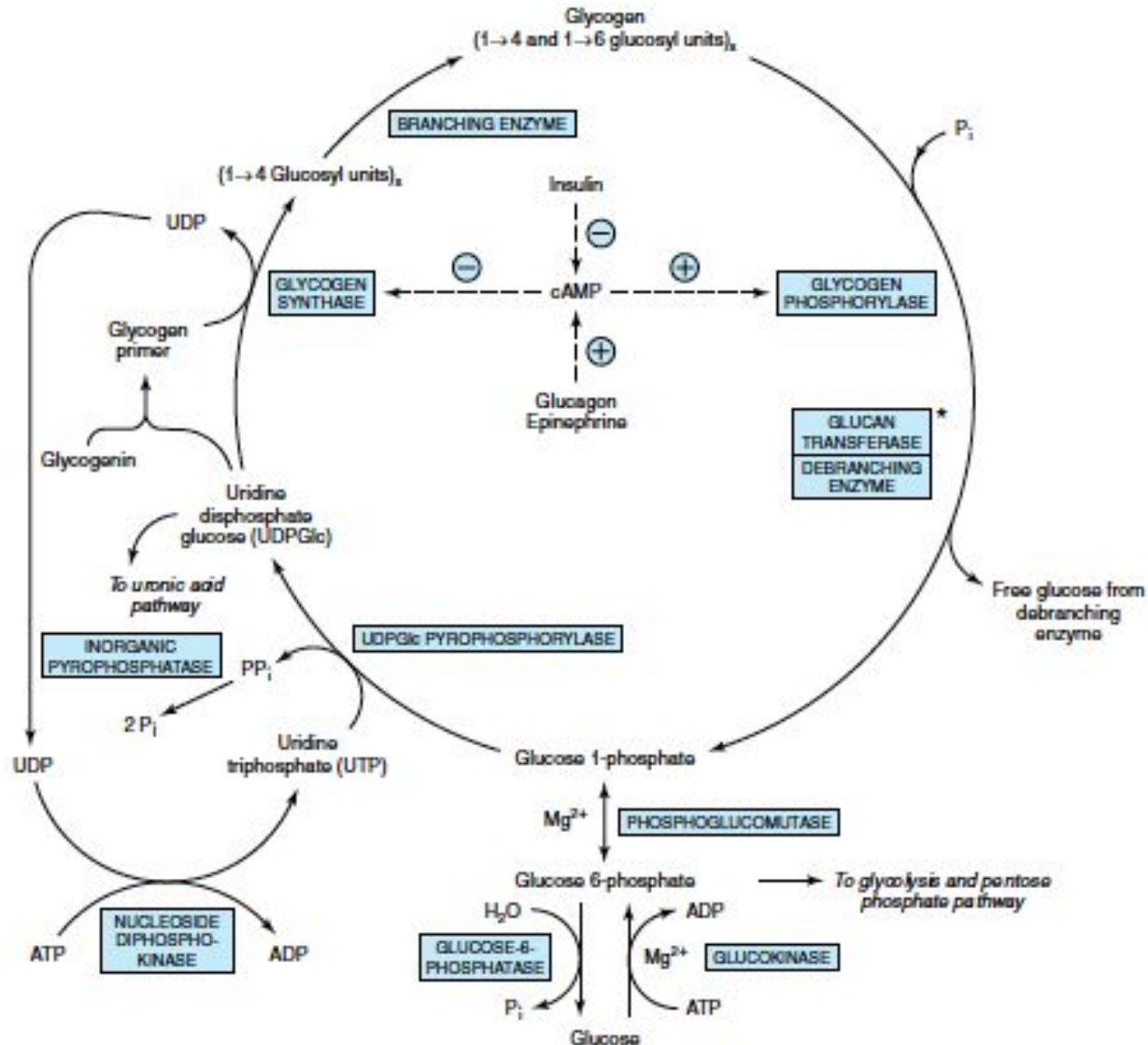


- Conversion of G1P (glucose-1-phosphate) to G6P (glucose-6-phosphate) through the enzyme phosphoglucomutase.

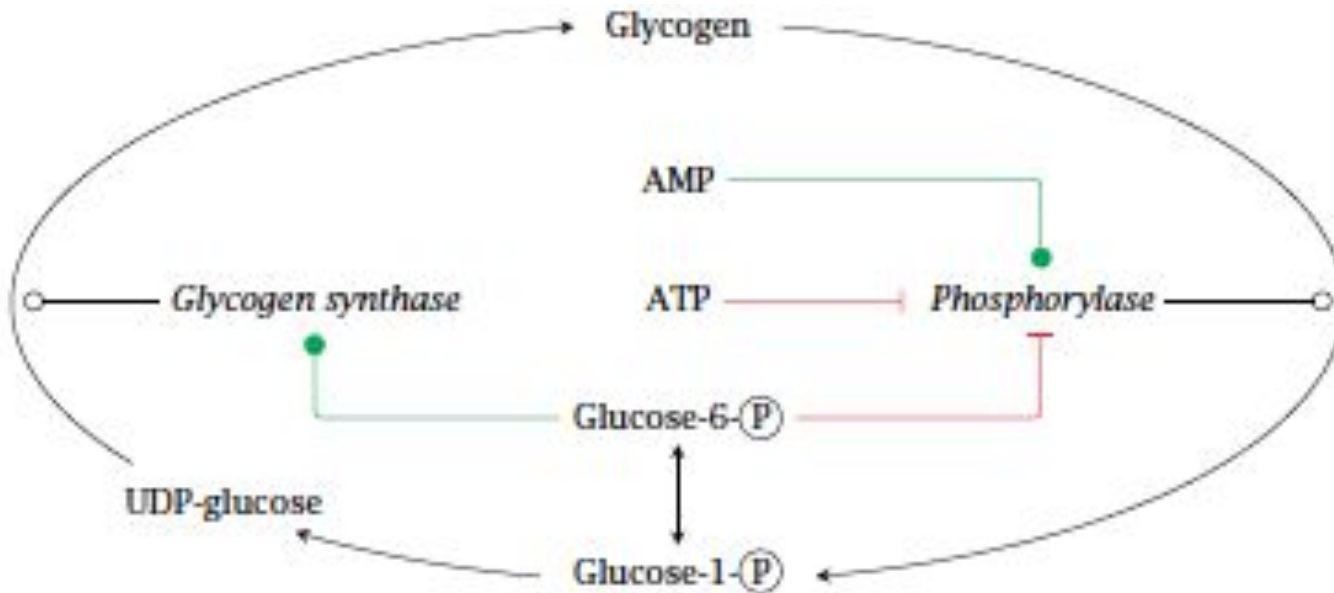
Alternate fates of glycogen-derived glucose 6-phosphate in muscle and liver.



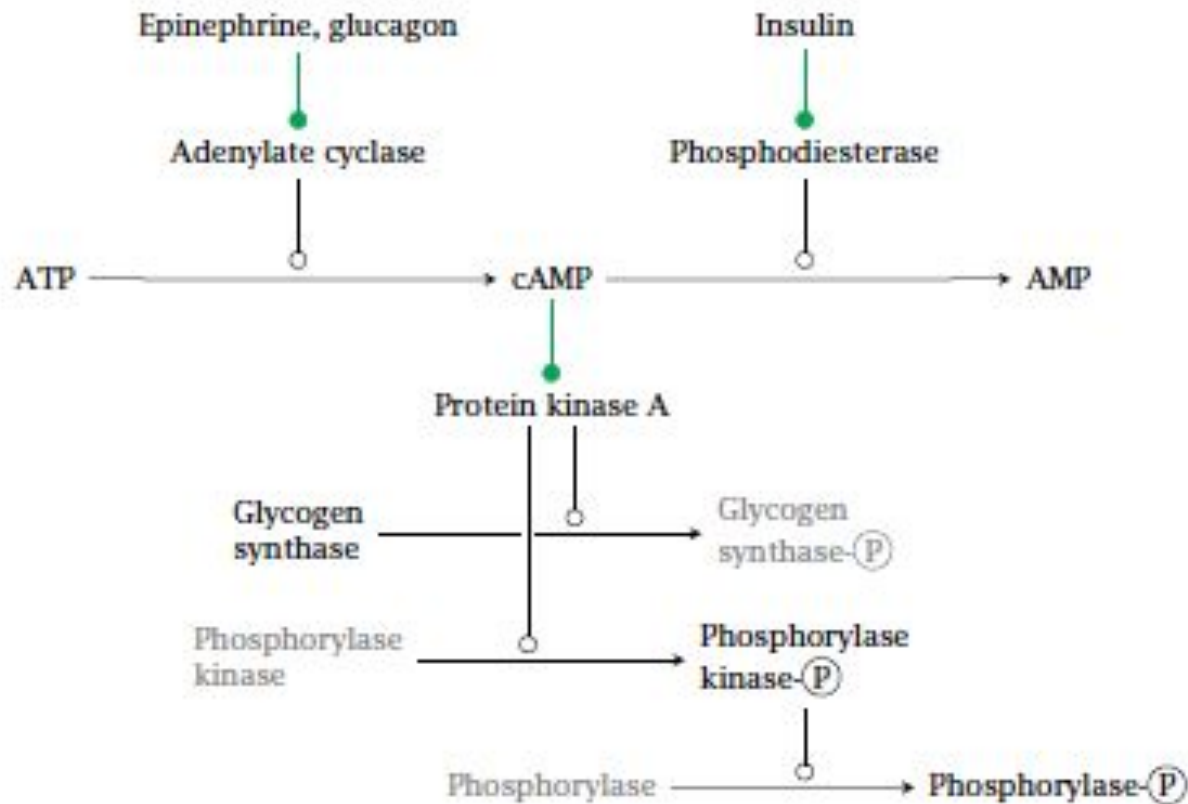
Summary of Glycogen metabolism



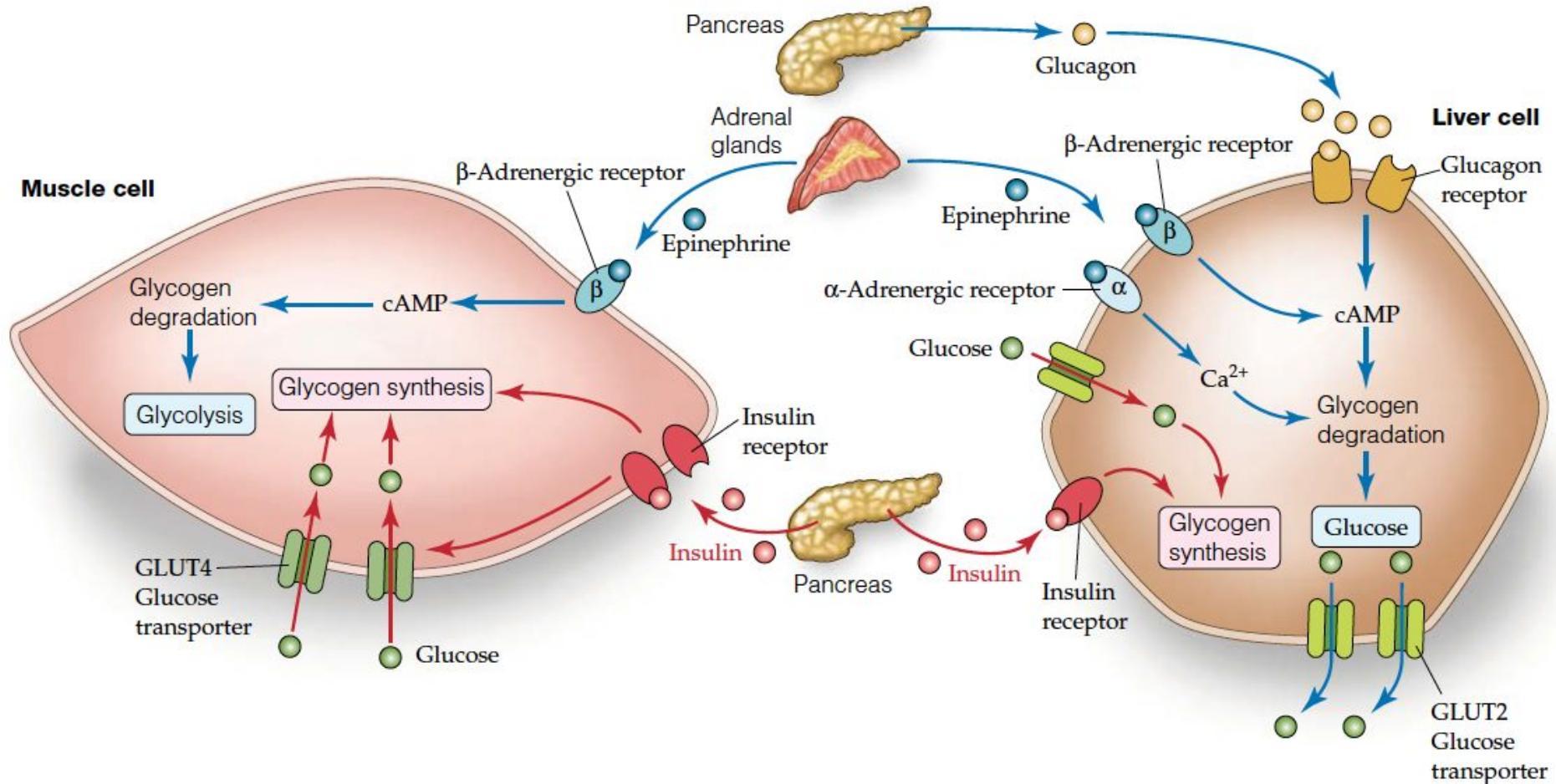
Regulation of glycogen metabolism



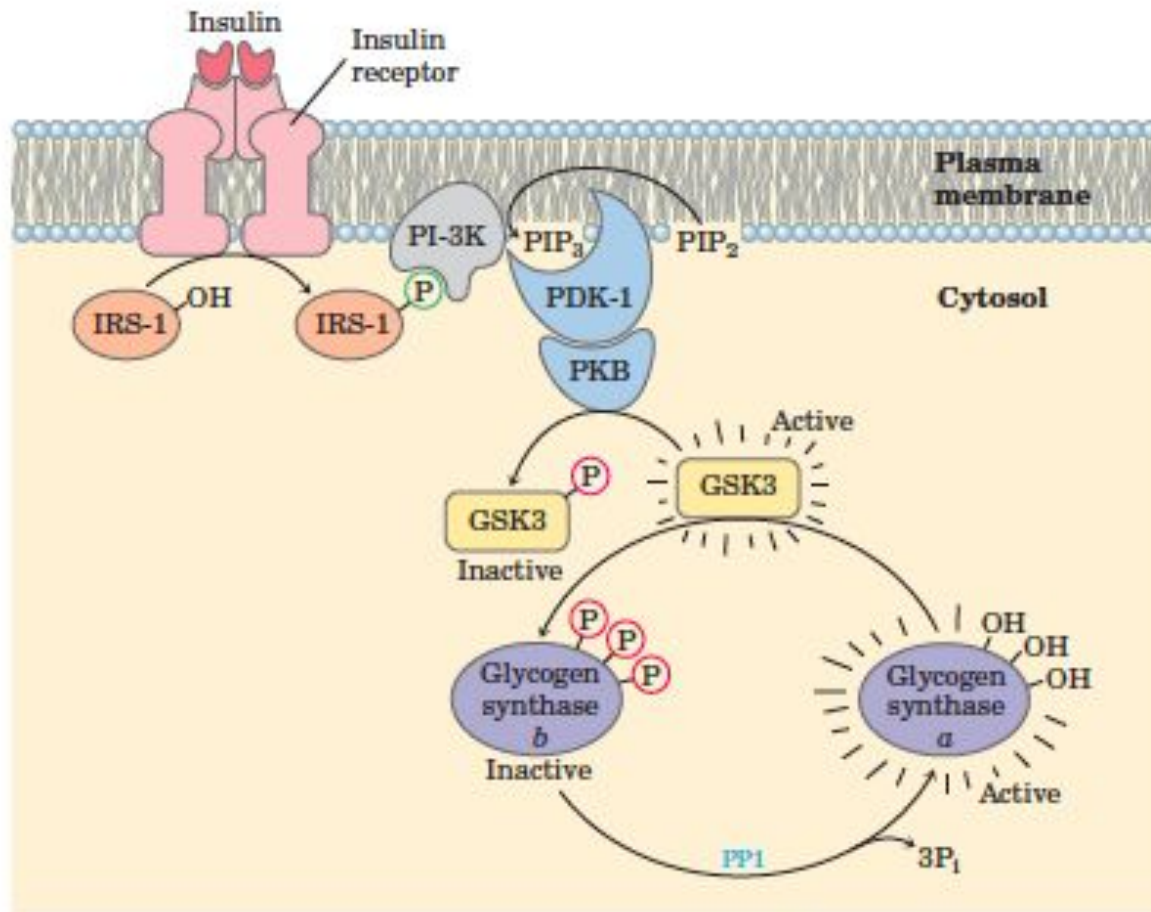
Hormonal control of glycogen metabolism



Hormonal control of glycogen metabolism



Insulin on (stimulates) glycogen synthesis

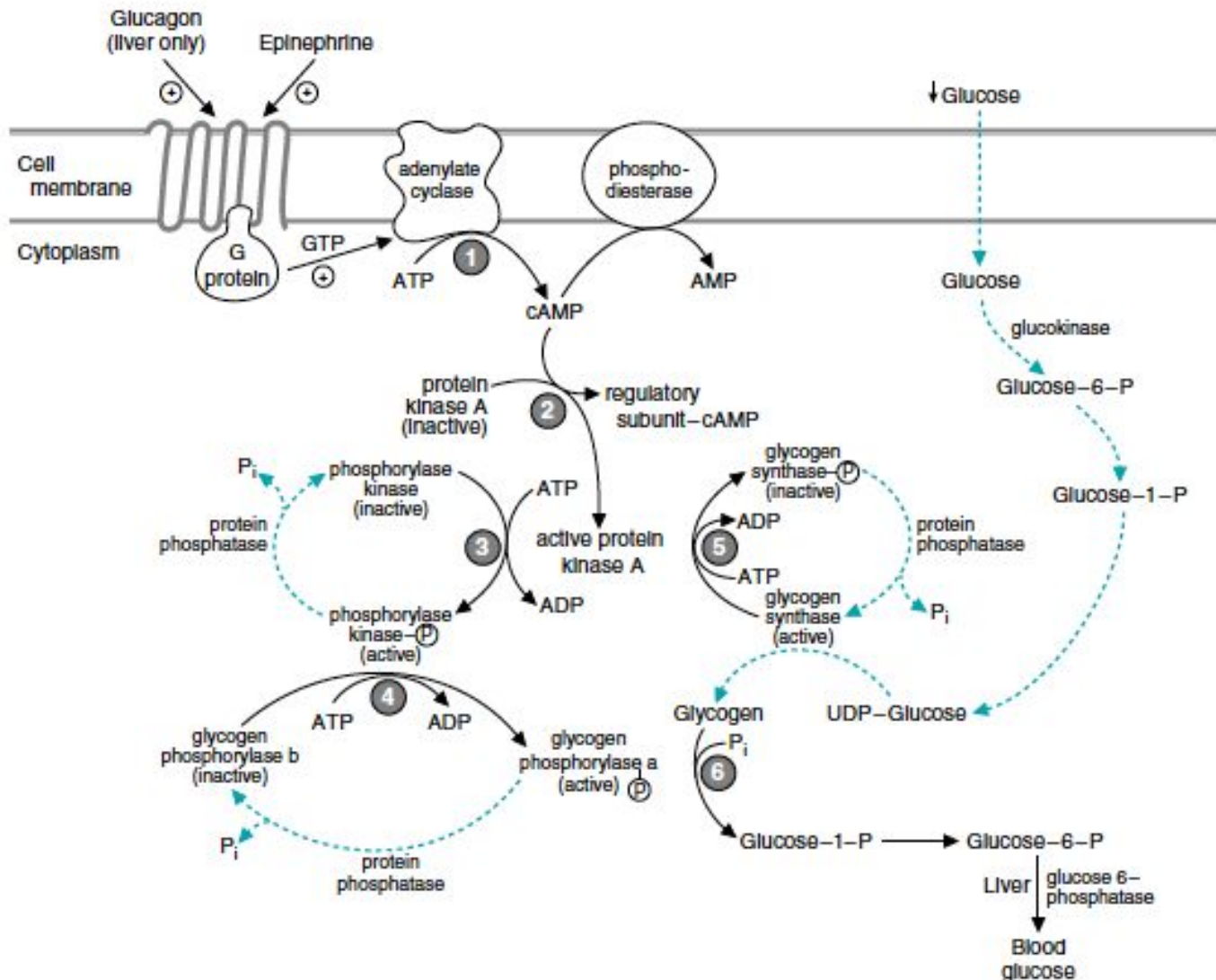


IRS-1, insulin receptor substrate-1; PI-3K, phosphatidylinositol 3-kinase; PIP₂, phosphatidylinositol 4,5-bisphosphate; PIP₃, phosphatidylinositol 3,4,5-trisphosphate. PDK-1, protein kinase; PKB, protein kinase; GSK3, glycogen synthase kinase 3; PP1, phosphoprotein phosphatase 1.

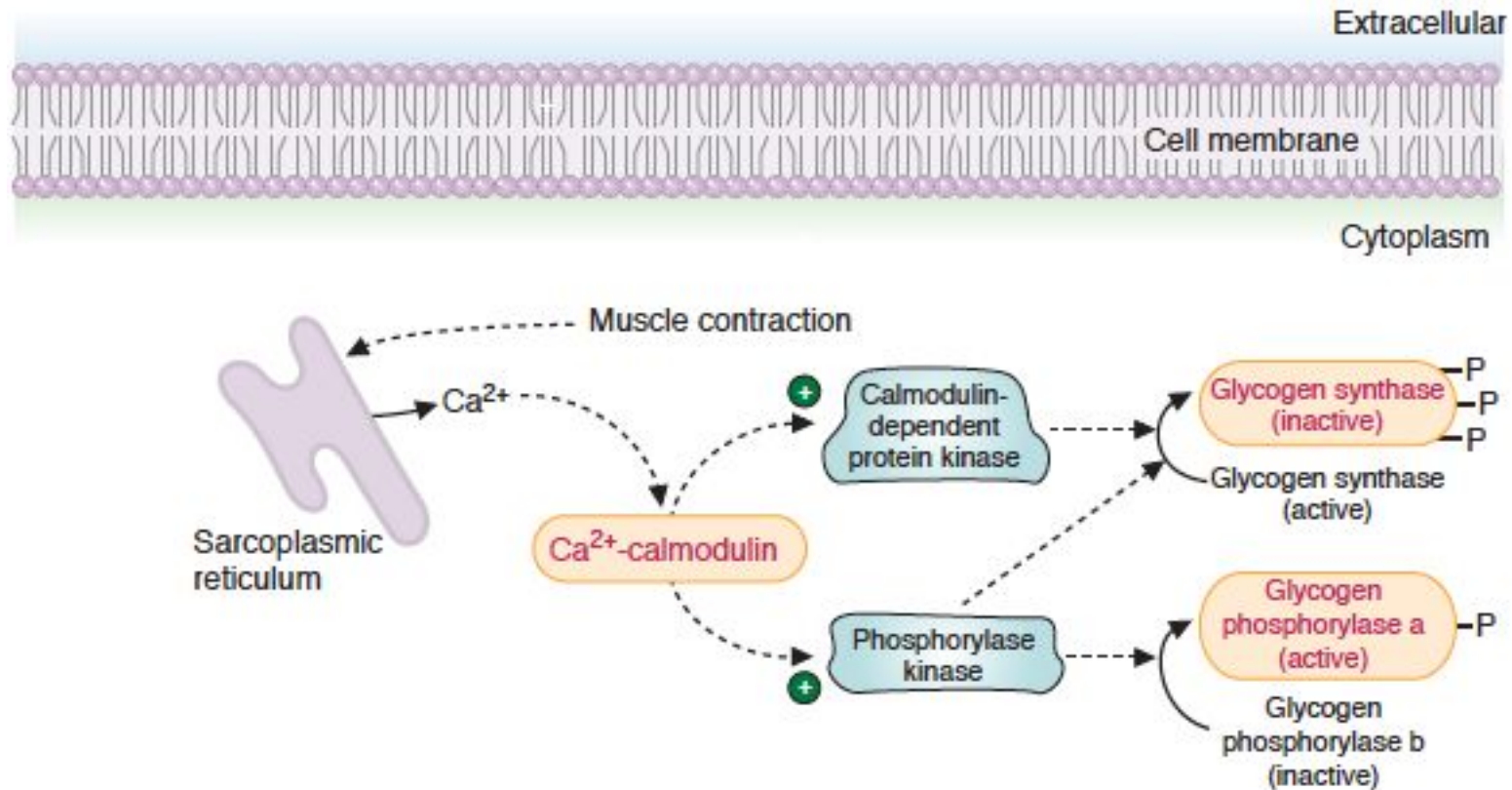
Epinephrine and Glucagon Stimulate Glycogen breakdown

- Muscle is responsive to epinephrine.
- Liver is responsive to glucagon and somewhat responsive to epinephrine.
- Both signal a cascade of molecular events leading to glycogen breakdown.
- Both utilize a G-protein-dependent signal-transduction pathway.

Regulation of glycogen synthesis and degradation in the liver



Regulation of glycogen synthesis and degradation by calcium in the muscle



Glycogen Storage Diseases (GSDs)

- Definition:
 - When the body is missing an enzyme or has a flawed enzyme and is not able to use glycogen the right way □ Glycogen storage disease (GSD).
 - Mostly a genetic condition
- Types:
 - At least 13 types of glycogen storage disease
- Frequency:
 - 1 in 20,000 to 25,000 babies
 - Nearly 90% of all patients with GSD have types I through IV
 - About 25% of patients with GSD are thought to have type I
- Causes:
 - Problem with the gene that has the instructions for making the enzyme that is missing or not working right.
 - The gene is passed down from parents to children.
 - In most cases, in order to have the GSD, a child must get a bad gene from both parents.

Glycogen Storage Diseases of Humans

Type (name)	Enzyme affected	Primary organ affected	Symptoms
Type 0	Glycogen synthase	Liver	Low blood glucose, high ketone bodies, early death
Type Ia (von Gierke's)	Glucose 6-phosphatase	Liver	Enlarged liver, kidney failure
Type Ib	Microsomal glucose 6-phosphate translocase	Liver	As in Ia; also high susceptibility to bacterial infections
Type Ic	Microsomal P _i transporter	Liver	As in Ia
Type II (Pompe's)	Lysosomal glucosidase	Skeletal and cardiac muscle	Infantile form: death by age 2; juvenile form: muscle defects (myopathy); adult form: as in muscular dystrophy
Type IIIa (Cori's or Forbes's)	Debranching enzyme	Liver, skeletal and cardiac muscle	Enlarged liver in infants; myopathy
Type IIIb	Liver debranching enzyme (muscle enzyme normal)	Liver	Enlarged liver in infants
Type IV (Andersen's)	Branching enzyme	Liver, skeletal muscle	Enlarged liver and spleen, myoglobin in urine
Type V (McArdle's)	Muscle phosphorylase	Skeletal muscle	Exercise-induced cramps and pain; myoglobin in urine
Type VI (Hers's)	Liver phosphorylase	Liver	Enlarged liver
Type VII (Tarui's)	Muscle PFK-1	Muscle, erythrocytes	As in V; also hemolytic anemia
Type VIb, VIII, or IX	Phosphorylase kinase	Liver, leukocytes, muscle	Enlarged liver
Type XI (Fanconi-Bickel)	Glucose transporter (GLUT2)	Liver	Failure to thrive, enlarged liver, rickets, kidney dysfunction

Classical types of GSDs

<i>Type/Name</i>	<i>Enzyme deficiency</i>
• Type I-von Gierke disease • Type II-Pompe's disease • Type III-Forbe's disease	Glucose-6-Pase Acid maltase Debranching enzyme (Limit dextrinosis)
• Type IV-Andersen's disease	Branching enzyme. (Amylopectinosis)
• Type V-McArdle's disease • Type VI-Her's disease	Muscle phosphorylase Liver phosphorylase

Symptoms of glycogen storage disease (GSD)

- Types 0, I, III, IV, VI and IX mostly affect liver and the most common symptoms are;
 - Enlarged liver
 - hypoglycemia, include sweating, tremor, drowsiness, confusion and sometimes seizures
- Types V and VII mostly affect the skeletal muscles and the most common symptoms are;
 - Muscle weakness and muscle cramps
- Other symptoms that may occur include:
 - Tiredness.
 - Very slow growth.
 - Obesity (being very overweight).
 - Problems with bleeding and blood clotting.
 - Kidney problems.
 - Low resistance to infections.
 - Breathing problems.
 - Heart problems.
 - Mouth sores.
 - Gout.

von Gierke disease (Type I-GSD)

- Clinical Manifestations with biochemical correlations:
 - As glucose-6-phosphatase enzyme is absent □ Breakdown of Liver glycogen↓ □ hypoglycaemia.
 - Energy demand↑ □ Fat oxidation↑ □ acetyl CoA↑ □ ketone bodies formation↑, cholesterol biosynthesis↑ □ xanthomas.
 - Persistent hypoglycaemia □ insulin secretion↓ □ protein synthesis↓ □ stunted growth and dwarfism.
 - Hypoglycaemia □ catecholamines↑ and glucagon↑ □ glycogenolysis in muscles↑ □ lactic acid (lactic acidosis).
 - Increased lactic acid in blood competes with uric acid excretion by kidneys □ increased blood uric acid↑ level □ Gout.
 - Normal type of glycogen accumulates in large quantities in liver, kidney and intestinal mucosal cells inhibiting their functions.

Indicators and diagnosis for GSDs

- Indicators:
 - A low blood glucose level.
 - An enlarged liver.
 - Lagging growth.
 - Abnormal blood tests.
- Diagnosis:
 - Family medical history
 - Because GSDs can run in families
 - Blood tests
 - To find out your blood glucose level and to see how your liver, kidneys and muscles are working.
 - Abdominal ultrasound
 - To see if your liver is enlarged.
 - Tissue biopsy
 - Testing a sample of tissue from a muscle or your liver to measure the level of glycogen or enzymes present.
 - Gene testing
 - To look for problems with the genes for different enzymes. Gene testing can confirm a GSD.

Management/ Treatment Option for GSDs

- Taking regular doses of uncooked cornstarch and/or nutrition supplements.
- Eating many small meals that are low in sugar.
- Continuous administration of carbohydrate solutions during the night.
- liver transplantation: for progressive liver failure, after a thorough evaluation.

Case file: Type 1 glycogen storage disorder

- A patient with type 1 glycogen storage disorder presents with symptoms of hepatomegaly and hypoglycemia.
- Biochemical changes:
 - Decreased activity of glucose-6-phosphatase.
 - Impaired glucose production and storage
 - Accumulation of glycogen in the liver.
 - Hyperlactemia and hypeuricemia
 - Increased ketone bodies production and Cholesterol synthesis.
- Diagnosis:
 - Genetic testing.
 - Blood analysis
 - Ultrasound tests.
- Management:
 - Involves dietary modifications, frequent meals, and monitoring blood glucose levels to prevent hypoglycemic episodes and liver complications.

Read more...



THANK YOU...